«You cannot step twice into the same river.» — *Heraclitus*

1. Karnaugh Maps

a) Generate the Function

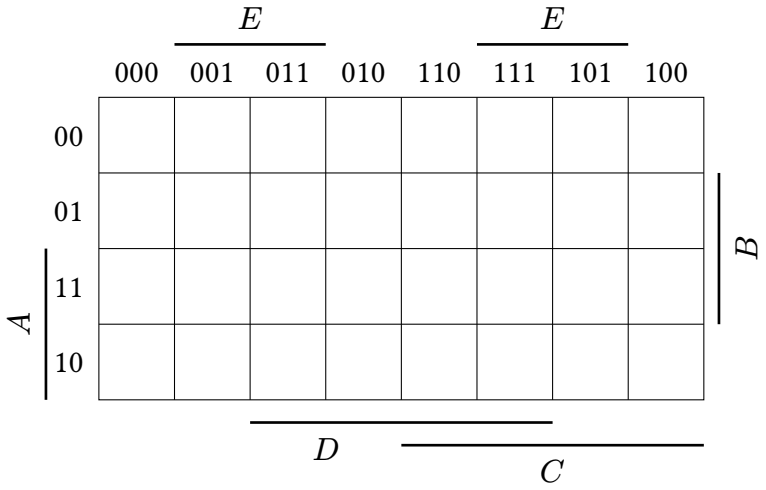
1. Compute[1] the SHA-256 *hash* h of string $s =$ «DM Fall 2025 HW4» (without quotes).
2. Split h into eight 32-bit blocks: $h = h_0 \parallel h_1 \parallel \ldots \parallel h_7$ where each h_i is 32 bits.
3. XOR all blocks together: $d = h_0 \oplus h_1 \oplus \cdots \oplus h_7$.
4. Apply the mask: $w = d \oplus$ `0x71be8976`, giving the 32-bit result $w = (w_0 w_1 \ldots w_{31})_2$.

The resulting bit string w encodes the 5-variable *boolean function* $f^{(5)}$: bit w_0 (MSB) represents $f(0,0,0,0,0)$, and bit w_{31} (LSB) represents $f(1,1,1,1,1)$.

Check: Hash h ends with …`01100010`; the value d (before masking) begins with `0000`…

b) Trying to master K-Map

Construct a 5-variable *Karnaugh map* for your function using the template below:





[1] You can use any programming language, tool or online service to compute SHA-256 hashes.

c) Find Minimal Forms

1. Find *minimal* DNF and CNF for f using your K-map.
2. Count all *prime implicants*. (PI).
3. If inputs where $A \wedge B \wedge C = 1$ are *don't-care conditions*[2] , how do the minimal forms change?

d) Analysis and Limitations

1. How many fewer *literals* does your minimal DNF use compared to full minterm expansion?
2. Which is more compact for your function: DNF or CNF? Why?

2. Boolean Function Analysis

Consider the following four functions[3]:

1. $f_1 = f^{(4)}_{47541}$
2. $f_2 = \sum m(1, 4, 5, 6, 8, 12, 13)$
3. $f_3 = f^{(4)}_{51011} \oplus f^{(4)}_{40389}$
4. $f_4 = A\overline{B}D + \overline{A}\,\overline{C}D + \overline{B}C\overline{D} + A\overline{C}D$

a) Boolean Function Decomposition

1. Decompose to SoP using *Shannon expansion* for function *4.* (without constructing the truth table).
2. Construct PoS for function *4.* from the known SoP.

b) K-Maps

Draw and fill the K-map for each function.

c) Prime Implicants

1. List all *implicants* of function *4.*
2. Using the K-map, find all *prime implicants* of function *4.* (Blake Canonical Form)[4]
3. Using the *Quine-McCluskey method*, find all *prime implicants* of function *4.* (Blake Canonical Form)
4. Determine which of them are *essential* (EPI).

d) Minimal DNF and CNF

1. Using *Petrick's method* (on the already obtained result of the Quine-McCluskey method) find the minimal DNF of function *4.*
2. Using *algebraic method* find the minimal CNF of function *4.*

e) Algebraic Normal Form

1. Construct the ANF for function *1.* using *Pascal's method*.
2. Construct the ANF for function *2.* using *triangle method*.
3. Construct the ANF for function *4.* using *K-map*.

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[2]Don't-care conditions are input combinations whose output values don't matter, allowing flexibility in minimization. They can be treated as either 0 or 1 to produce smaller expressions.

[3]Notation: $f^{(n)}_k$ is the k-th Boolean function of n variables, where k is the decimal truth table value, which, represented in binary as $(f_0 f_1 \ldots f_{2^n-1})_2$, corresponds to values of f: MSB f_0 is $f(0, \ldots, 0)$, LSB f_{2^n-1} is $f(1, \ldots, 1)$.

[4]Reduced DNF - Blake Canonical Form

f) Zhegalkin Polynomial

1. Identify the *algebraic degree* of each function from from part (e).
2. Which function has the highest degree? Is it *maximal* for this number of variables?
3. Prove that $\deg(f \oplus g) \leq \max(\deg(f), \deg(g))$. Find an example with strict inequality.

3. Reed–Muller Codes

(optional, +1 extra point)

In 1971, Mariner 9 transmitted the first close-up Mars images using Reed–Muller codes. These error-correcting codes are built from Boolean functions with restricted algebraic degree.

The Reed–Muller code $\mathrm{RM}(r, m)$ consists of all Boolean functions $f : \mathbb{F}_2^m \to \mathbb{F}_2$ with $\deg(f) \leq r$.

Parameters: length $n = 2^m$, dimension $k = \sum_{i=0}^{r} \binom{m}{i}$, minimum *Hamming distance*[5] $d = 2^{m-r}$.

Generator matrix construction: Rows correspond to *monomials* of degree $\leq r$. Columns correspond to inputs $(x_1, \ldots, x_m) \in \mathbb{F}_2^m$. Entry (i, j) is the value of monomial i evaluated at input j.

Example: $\mathrm{RM}(1, 2)$ has monomials $\{1, x_1, x_2\}$ and inputs $\{(0,0), (0,1), (1,0), (1,1)\}$:

$$G = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

Message $(1, 0, 1)$ encodes to $(1, 0, 1, 0)$ via $\boldsymbol{c} = \boldsymbol{u}G$.

a) Code Construction

Consider $\mathrm{RM}(1, 3)$:

1. Compute the code parameters: length n, dimension k, minimum distance d, and error correction capability $t = \left\lfloor \frac{d-1}{2} \right\rfloor$.
2. Construct the 4×8 generator matrix G. Rows correspond to monomials $\{1, x_1, x_2, x_3\}$. Columns are ordered: $(0,0,0), (0,0,1), \ldots, (1,1,1)$, where column i (for $i = 1, \ldots, 8$) corresponds to the binary representation of $i - 1$.
3. Encode the message $\boldsymbol{u} = (1, 0, 1, 1)$ into codeword $\boldsymbol{c} = \boldsymbol{u}G$.

b) Single-Error Correction

1. Flip *one* bit at position $i \in \{1, \ldots, 8\}$ of your codeword $\boldsymbol{c}$ to simulate a transmission error, obtaining received word $\boldsymbol{r} = (r_1, \ldots, r_8)$.

2. Apply majority-logic decoding to recover the ANF coefficients (a_0, a_1, a_2, a_3):
   - For each a_j where $j > 0$: identify all pairs of positions (p, q) whose input vectors differ only in variable x_j Each pair gives a vote $v = r_p \oplus r_q$. Set $a_j = \text{majority}(v_1, v_2, \ldots)$.
   - For a_0: after determining a_1, a_2, a_3, compute $a_0 = r_1 \oplus (a_1 \cdot 0) \oplus (a_2 \cdot 0) \oplus (a_3 \cdot 0) = r_1$, since position 1 corresponds to input $(0, 0, 0)$.[6]

[5] The *Hamming distance* between two codewords is the number of positions in which they differ.

3. Verify that the decoded message (a_0, a_1, a_2, a_3) equals your original $\boldsymbol{u} = (1, 0, 1, 1)$.

4. Show the detailed computation for coefficient a_1: list all pairs differing in x_1, compute each XOR vote, and determine the majority.

[6] In case of tied votes, choose 0 as the default value. Note that with a single error, ties should not occur for $\mathrm{RM}(1, 3)$.